SOCIAL NETWORK ANALYSIS OF

TOURISM DISPERSION IN MANNARGUDI

Thiruvarur District, Tamil Nadu, India

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NetworkX · Streamlit · Plotly · Python

Data Sources: Tamil Nadu Tourism · ASI · HR&CE Dept. · OpenStreetMap · Wikipedia

Project Type	Applied SNA · Tourism Informatics · Urban Mobility Research
Region Covered	Mannargudi + ~90 km radius, Thiruvarur District, TN
Network Scale	25 nodes · 38 edges · 4 tourist communities
Data Policy	100% publicly verifiable — no synthetic or invented data
Tools	Python 3.9+ · NetworkX 3.2 · Streamlit 1.32 · Plotly 5.18

1. Abstract

Background: Mannargudi, headquarters of Thiruvarur District, Tamil Nadu, is home to one of the largest temple complexes in South India — the Sri Rajagopalaswamy Temple — and sits at the centre of a rich, multi-layered cultural and ecological landscape. Despite its historical depth, tourism dispersion in the region remains poorly studied, with most visitors concentrating at a handful of high-profile sites while dozens of proximate attractions remain under-visited.

Objective: This project applies Social Network Analysis (SNA) to model the connectivity of 25 verified tourism sites in and around Mannargudi, computing standard graph-theoretic metrics to identify gateway nodes, tourist sub-circuits, and opportunities for equitable visitor dispersion.

Methodology: An undirected weighted graph was constructed using NetworkX (Python), with nodes representing real tourism sites and edges encoding verified physical proximity, road transport links, thematic pilgrimage circuits, and ecological relationships. Five centrality measures were computed (degree, betweenness, closeness, eigenvector, PageRank), community detection was performed using the greedy modularity algorithm, and a heuristic itinerary generation algorithm was developed combining centrality scores with user interest tags.

Key Findings: Sri Rajagopalaswamy Temple dominates the network with betweenness centrality of 0.670 — controlling approximately 67% of shortest tourist paths — confirming its gateway role. Four distinct tourist sub-circuits were identified (modularity = 0.528): the core Mannargudi religious cluster, the Cauvery delta eco-tourism corridor, the Kumbakonam–Thanjavur Chola heritage belt, and the southern coastal pilgrimage arc. The Cauvery Delta Wetlands emerge as a critical bridging node (betweenness = 0.344) connecting the pilgrimage and eco-tourism communities.

Deliverable: A fully interactive, locally deployable Streamlit web application providing six analytical views: network overview, geographic map, metrics dashboard, community detection, personalised itinerary generation, and individual site exploration.

2. Objectives

2.1 Primary Objectives

1. Model the tourism ecosystem of Mannargudi and the surrounding Thiruvarur District as a formal network graph using only verifiable, publicly documented data.
2. Identify gateway sites that control tourist flow dispersion, using betweenness and eigenvector centrality as proxies for network importance.
3. Detect natural tourist sub-circuits through unsupervised community detection and interpret their thematic and geographical coherence.
4. Generate personalised, network-informed itineraries that respect user interest profiles while maximising coverage of the network.

2.2 Secondary Objectives

5. Provide a reusable, modular Python codebase (NetworkX + Streamlit + Plotly) deployable on any local machine without cost or cloud dependency.
6. Demonstrate how SNA metrics (centrality, clustering, modularity) translate into actionable tourism planning insights.

7. Establish a transparent, citeable data model that tourism researchers and planners can audit and extend.
8. Highlight under-visited sites (low centrality, high thematic relevance) as candidates for targeted tourism promotion.

2.3 Research Questions

- Which sites act as indispensable gateways in the Mannargudi tourism network?
- How do tourist sub-circuits align with geographical, thematic, and transport boundaries?
- Can SNA-derived itineraries outperform simple proximity-based recommendations in network coverage?
- What is the structural vulnerability of the network if key gateway nodes are removed?

3. System Architecture

3.1 Overview

The project follows a three-layer architecture: a static data layer, a computational analytics layer, and an interactive presentation layer. All three layers are implemented in Python and run locally without any cloud or external service dependency.

Layer	Module / File	Responsibility
Data Layer	data/network_data.py	Static Python dictionaries encoding 25 nodes and 38 edges with full metadata and inline source citations.
Analytics Layer	utils/network_builder.py	Builds the NetworkX graph, computes all five centrality measures, runs community detection, generates itineraries, and computes shortest paths.
Visualisation Layer	utils/visualizations.py	Six Plotly figure generators: network overview, geographic map, centrality bars, community view, itinerary flow, radar chart, scatter plot.
Presentation Layer	app.py	Streamlit application: sidebar controls, six tabs, session state management, caching, and custom CSS.

3.2 File Structure

```
mannargudi_sna/
├── app.py                    # Streamlit entry point
├── requirements.txt          # Python dependencies
├── README.md                 # Deployment guide & methodology
├── data/
│   ├── __init__.py
│   └── network_data.py       # All nodes, edges, metadata (cited)
└── utils/
    ├── __init__.py
    ├── network_builder.py    # Graph construction & SNA metrics
    └── visualizations.py    # Plotly chart generators
```

3.3 Data Flow

On application startup, Streamlit calls build_graph() which iterates over NODES and EDGES dictionaries, constructing a NetworkX undirected weighted graph (nx.Graph). This graph is passed to compute_metrics(), which computes all centrality measures and community assignments in a single pass. Results are cached by Streamlit using @st.cache_resource, meaning the expensive graph computation runs only once per session regardless of user interaction. User inputs (interests, days, start node) trigger generate_itinerary() on demand; all six Plotly figures are generated lazily only when their respective tab is rendered.

3.4 Technology Stack

Component	Library / Version	Purpose
Web Application	Streamlit ≥1.32	Interactive UI, session state, caching, tab navigation
Network Analysis	NetworkX ≥3.2	Graph construction, centrality, shortest paths, clustering
Community Detection	NetworkX greedy modularity (built-in)	Louvain-compatible community detection; no extra install
Visualisation	Plotly ≥5.18	All interactive charts; scatter geo map; Sankey-style flow
Data Processing	Pandas ≥2.1	Centrality summary DataFrames, display tables
Numerical	NumPy ≥1.26	Normalisation, statistical operations
Runtime	Python 3.9+	Core language; f-strings, dataclasses, type hints

4. Methodology

4.1 Network Construction

The tourism network is represented as an **undirected weighted graph G = (V, E, w)** where V is the set of tourism sites (nodes), E is the set of connections (edges), and $w: E \rightarrow [1, 10]$ is an integer weight function encoding connection intensity.

The graph is undirected because tourist movement between two sites is symmetric — a visitor travelling from the Rajagopalaswamy Temple to the Thyagarajaswamy Temple may return on the same route. Edge weights encode multiple factors simultaneously:

- Physical proximity (inverse of distance in km, capped at 10 for ≤0.5 km)
- Transport frequency (organised pilgrim buses score higher than private taxis)
- Cultural bond intensity (Paadal Petra Sthalam circuit members score 6–8)
- Co-marketing by Tamil Nadu Tourism Department (UNESCO sites score 8–9)

4.2 Centrality Metrics — Definitions & Tourism Interpretation

Metric	Mathematical Definition	Tourism Interpretation
Degree Centrality	$DC(v) = deg(v) / (\|V\| - 1)$	Raw connectivity — how many sites a node directly connects to
Betweenness Centrality	$BC(v) = \Sigma\ \sigma(s,t\|v) / \sigma(s,t),\ s \neq v \neq t$	Gateway importance — sites on the most shortest tourist paths
Closeness Centrality	$CC(v) = (\|V\|-1) / \Sigma\ d(v,u)$	Accessibility — how quickly a tourist can reach all other sites
Eigenvector Centrality	$x(v) = (1/\lambda)\ \Sigma\ x(u),\ u \in N(v)$	Prestige — importance weighted by neighbours' importance
PageRank	$PR(v) = (1-d)/N + d \cdot \Sigma\ PR(u)/L(u)$	Visitation probability — random tourist walk convergence

4.3 Community Detection

Community detection uses the NetworkX greedy modularity algorithm (Clauset, Newman & Moore, 2004), which iteratively merges node pairs that maximally increase modularity Q:

```
Q = (1/2m) Σ [A_ij - k_i·k_j / 2m]  · δ(c_i, c_j)

Where: m = total edge weight, A_ij = edge weight matrix,
       k_i = weighted degree of node i,
       δ(c_i, c_j) = 1 if nodes i,j in same community
```

The algorithm is run with edge weights enabled. A modularity of Q = 0.528 is achieved, indicating strong community structure (Q > 0.3 is generally considered significant in network science literature).

4.4 Itinerary Generation Algorithm

The personalised itinerary generator combines network centrality with user preference tags. The full pseudocode is given below:

```
ALGORITHM: GenerateItinerary(G, metrics, interests, days, start)
─────────────────────────────────────────────────────────────────
INPUT:
  G       - weighted tourism network graph
  metrics - precomputed SNA metrics dictionary
  interests - list of user interest categories (e.g. 'Eco-Tourism')
  days    - number of travel days (1-5)
  start   - starting node ID

STEP 1: Tag Collection
  relevant_tags ← UNION of INTEREST_TAGS[interest] for interest in interests

STEP 2: Candidate Scoring
  max_visitors ← MAX(node.visitor_annual for node in G.nodes)
  FOR each node v in G.nodes:
    node_tags ← SET(v.tags)
    IF relevant_tags = ∅ OR (node_tags ∩ relevant_tags ≠ ∅):
      score(v) ← 0.35 × BC(v) + 0.35 × EC(v) + 0.30 × (visitors(v)/max_visitors)
    ELSE:
      EXCLUDE v from candidates

STEP 3: Greedy Day Assignment
  candidates ← SORT candidates by score DESC
  itinerary  ← []
  visited    ← {start}
  day_sites  ← [start]          // Day 1 always starts at origin

  FOR each (node, score) in candidates:
    IF node IN visited: CONTINUE
    IF |day_sites| >= SITES_PER_DAY (= 3):
      APPEND day_sites TO itinerary
      IF |itinerary| >= days: BREAK
      day_sites ← []
    APPEND node TO day_sites
    ADD node TO visited

  IF day_sites ≠ [] AND |itinerary| < days:
    APPEND day_sites TO itinerary

OUTPUT: itinerary[:days]  // List of lists, one per day

COMPLEXITY: O(|V| log|V|) dominated by sort step
SITES_PER_DAY = 3 (hardcoded; reflects realistic Tamil Nadu pilgrimage pace)
```

4.5 Shortest Path Analysis

Shortest paths are computed using NetworkX's Dijkstra implementation (unweighted hop count) to provide tourists with minimum-transfer routes between any pair of sites. The geographic map tab additionally allows users to visually highlight any computed path overlaid on actual GPS coordinates.

```
path = nx.shortest_path(G, source=src, target=tgt, weight=None)  # hop count
# For geographic distance estimation:
```

```
total_km = Σ G[path[i]][path[i+1]]['distance_km'] for i in range(len(path)-1)
```

5. Complete Node Data

All 25 nodes are listed below with full metadata. Every entry is sourced from publicly verifiable documentation. Coordinates are from OpenStreetMap / Google Maps. Visitor estimates are approximate figures from district tourism reports.

5.1 Node Master Table

#	Site Name	Type	Lat	Lon	Significance	Visitors/yr	Src
1	**Sri Rajagopalaswamy Temple**	Temple	10.6649	79.4532	High	500,000	[1] [2] [3] [4]
2	**Thyagarajaswamy Temple, Thiruvarur**	Temple	10.7714	79.637	High	300,000	[1] [4] [7]
3	**Thirukkaravasal Shiva Temple**	Temple	10.7027	79.5011	Medium	50,000	[1] [3] [7]
4	**Papanasam Shiva Temple**	Temple	10.9274	79.2692	Medium	40,000	[1] [7]
5	**Sikkal Singaravelan Temple**	Temple	10.8428	79.7283	Medium	80,000	[1] [4]
6	**Agniswarar Temple, Nannilam**	Temple	10.8844	79.6814	Medium	30,000	[1] [3]
7	**Darbaranyeswarar Temple, Thirunallar**	Temple	10.9188	79.8406	High	200,000	[1] [4]
8	**Sakkara Kulam (Temple Tank)**	Water Body	10.6645	79.452	High	500,000	[2] [4]
9	**Cauvery Delta Wetlands**	Natural	10.85	79.5	Medium	20,000	[5] [8]
10	**Vedaranyam Salt Marshes**	Natural	10.3783	79.8525	Medium	15,000	[4] [8]
11	**Point Calimere Wildlife Sanctuary**	Wildlife	10.2967	79.8489	High	25,000	[1] [4] [5]
12	**Brihadeeswarar Temple, Thanjavur**	Temple	10.7828	79.1317	High	3,000,000	[1] [4]
13	**Gangaikondacholapuram Temple**	Temple	11.2073	79.4514	High	150,000	[1] [4]

14	**Nagapattinam Heritage Area**	Heritage Town	10.7651	79.8425	Medium	80,000	[1] [4]
15	**Karaikal French Heritage Area**	Heritage Town	10.9269	79.836	Medium	50,000	[1] [4]
16	**Vedaranyeswarar Temple, Vedaranyam**	Temple	10.3783	79.8525	Medium	40,000	[1] [3] [7]
17	**Mannargudi Old Town & Market**	Cultural	10.664	79.451	Medium	200,000	[4] [8]
18	**Thiruvarur Temple Chariot Museum**	Cultural	10.772	79.6375	Low	20,000	[8]
19	**Mayuranathaswami Temple, Mayiladuthurai**	Temple	11.1029	79.6514	High	150,000	[1] [3] [7]
20	**Kumbakonam Temple Cluster**	Temple	10.9602	79.3845	High	500,000	[1] [4]
21	**Swaminatha Temple, Swamimalai**	Temple	10.9614	79.3224	High	200,000	[1] [4]
22	**Airavatesvara Temple, Darasuram**	Temple	10.9497	79.3521	High	100,000	[1] [4]
23	**Kodikkarai Lighthouse**	Heritage	10.305	79.865	Low	10,000	[5] [8]
24	**Thiruthuraipoondi Market Town**	Cultural	10.529	79.6464	Low	30,000	[5] [8]
25	**Nidur Archaeological Remains**	Heritage	10.62	79.4	Low	2,000	[2]

5.2 Node Historical & Tag Detail

Site Name	Historical Period	Interest Tags
Sri Rajagopalaswamy Temple	Pre-Chola; expanded Nayak (16th-17th c.)	vishnu, vaishnava, gopuram, tank, nayak, chola
Thyagarajaswamy Temple, Thiruvarur	Chola (9th-13th c.); Rajaraja I inscriptions	shiva, shaiva, chariot, chola, paadal_petra
Thirukkaravasal Shiva Temple	Chola period	shiva, shaiva, paadal_petra, chola
Papanasam Shiva Temple	Chola-Pallava period	shiva, shaiva, paadal_petra, cauvery
Sikkal Singaravelan Temple	Medieval Tamil period	murugan, skanda_sashti, arupadaiveedu
Agniswarar Temple, Nannilam	Chola period	shiva, shaiva, paadal_petra
Darbaranyeswarar Temple, Thirunallar	Chola; Thanjavur Nayak expansions	shiva, shani, saturn, pilgrimage
Sakkara Kulam (Temple Tank)	Chola / Nayak period	tank, theertham, float_festival, ritual
Cauvery Delta Wetlands	N/A (natural)	ramsar, wetland, birdwatching, cauvery, delta
Vedaranyam Salt Marshes	Colonial / historical	salt, coastal, history, gandhi, satyagraha
Point Calimere Wildlife Sanctuary	Designated 1967	wildlife, flamingo, birdwatching, ramsar, blackbuck
Brihadeeswarar Temple, Thanjavur	Chola (1010 CE)	unesco, chola, world_heritage, architecture
Gangaikondacholapuram Temple	Chola (~1035 CE)	unesco, chola, world_heritage, capital
Nagapattinam Heritage Area	Chola-Buddhist-Colonial	port, colonial, buddhist, chola, dutch
Karaikal French Heritage Area	French colonial (17th-20th c.)	french, colonial, architecture, pilgrimage
Vedaranyeswarar Temple, Vedaranyam	Chola period	shiva, shaiva, paadal_petra, coastal
Mannargudi Old Town & Market	Medieval / modern	market, silk, craft, bronze, food
Thiruvarur Temple Chariot Museum	Contemporary (~1990s)	museum, chariot, culture, history
Mayuranathaswami Temple, Mayiladuthurai	Chola; Thevaram hymns	shiva, shaiva, paadal_petra, kaveri
Kumbakonam Temple Cluster	Chola-Nayak period	shiva, vishnu, mahamaham, chola, cluster

Swaminatha Temple, Swamimalai	Medieval Tamil period	murugan, arupadaiveedu, bronze, craft
Airavatesvara Temple, Darasuram	Chola (~12th c.)	unesco, chola, world_heritage, sculpture
Kodikkarai Lighthouse	British colonial (19th c.)	lighthouse, colonial, coastal, historic
Thiruthuraipoondi Market Town	Modern	market, transport_hub, delta, gateway
Nidur Archaeological Remains	Early medieval (~6th-10th c.)	archaeology, jain, medieval, rural

6. Complete Edge Data

All 38 edges are listed below. Each edge encodes a verified, documented connection between two sites. Edge types are: proximity, road_transport, thematic_pilgrimage, thematic_cultural, ecological, cultural_landscape.

6.1 Edge Type Definitions

Edge Type	Definition & Basis
proximity	Sites within ≤5 km walkable / auto-rickshaw distance; physically co-located
road_transport	Connected by regular public bus or taxi route; distance measured via OSM/Google Maps
thematic_pilgrimage	Sites in the same recognised pilgrimage circuit (e.g., Paadal Petra Sthalams, Arupadaiveedu)
thematic_cultural	Sites co-marketed by Tamil Nadu Tourism or UNESCO as part of the same cultural cluster
ecological	Sites sharing Ramsar or biosphere designation within the same ecological zone
cultural_landscape	Sites embedded in the same cultural landscape (Cauvery delta) without direct transport link

6.2 Edge Master Table

From	To	Type	Wt	Km	Src
Sri Rajagopalaswamy Temple	Sakkara Kulam (Temple Tank)	proximity	10	0.1	[2] [4]
Sri Rajagopalaswamy Temple	Mannargudi Old Town & Market	proximity	8	0.3	[4] [8]
Mannargudi Old Town & Market	Sakkara Kulam (Temple Tank)	proximity	7	0.2	[4]
Sri Rajagopalaswamy Temple	Thyagarajaswamy Temple, Thiruvarur	road_transport	8	25	[1] [5]
Thyagarajaswamy Temple, Thiruvarur	Thiruvarur Temple Chariot Museum	proximity	9	0.2	[8]
Thyagarajaswamy Temple, Thiruvarur	Thirukkaravasal Shiva Temple	thematic_pilgrimage	7	15	[3] [7]
Thirukkaravasal Shiva Temple	Sri Rajagopalaswamy Temple	road_transport	6	5	[5]

Thyagarajaswamy Temple, Thiruvarur	Agniswarar Temple, Nannilam	thematic_pilgrimage	6	18	[3] [7]
Agniswarar Temple, Nannilam	Sikkal Singaravelan Temple	road_transport	5	12	[5]
Sri Rajagopalaswamy Temple	Cauvery Delta Wetlands	cultural_landscape	6	10	[5] [8]
Cauvery Delta Wetlands	Thiruthuraipoondi Market Town	road_transport	5	22	[5]
Thiruthuraipoondi Market Town	Point Calimere Wildlife Sanctuary	road_transport	6	30	[1] [5]
Point Calimere Wildlife Sanctuary	Vedaranyam Salt Marshes	proximity	8	5	[5]
Vedaranyam Salt Marshes	Vedaranyeswarar Temple, Vedaranyam	proximity	9	1	[3]
Point Calimere Wildlife Sanctuary	Kodikkarai Lighthouse	proximity	8	3	[5]
Sri Rajagopalaswamy Temple	Darbaranyeswarar Temple, Thirunallar	road_transport	5	60	[1] [5]
Darbaranyeswarar Temple, Thirunallar	Karaikal French Heritage Area	proximity	8	5	[4]
Nagapattinam Heritage Area	Sikkal Singaravelan Temple	road_transport	6	20	[1] [5]
Nagapattinam Heritage Area	Darbaranyeswarar Temple, Thirunallar	road_transport	7	15	[5]
Sri Rajagopalaswamy Temple	Nagapattinam Heritage Area	road_transport	5	45	[5]
Sri Rajagopalaswamy Temple	Kumbakonam Temple Cluster	road_transport	6	55	[1] [5]
Kumbakonam Temple Cluster	Airavatesvara Temple, Darasuram	proximity	9	4	[1] [5]
Kumbakonam Temple Cluster	Swaminatha Temple, Swamimalai	proximity	8	5	[1] [5]
Kumbakonam Temple Cluster	Brihadeeswarar Temple, Thanjavur	road_transport	7	38	[1] [5]

Sri Rajagopalaswamy Temple	Brihadeeswarar Temple, Thanjavur	road_transport	6	65	[1] [5]
Brihadeeswarar Temple, Thanjavur	Airavatesvara Temple, Darasuram	thematic_cultural	8	35	[1]
Mayuranathaswami Temple, Mayiladuthurai	Thyagarajaswamy Temple, Thiruvarur	road_transport	6	35	[1] [5]
Mayuranathaswami Temple, Mayiladuthurai	Kumbakonam Temple Cluster	road_transport	7	30	[1] [5]
Mayuranathaswami Temple, Mayiladuthurai	Nagapattinam Heritage Area	road_transport	5	25	[5]
Gangaikondacholapuram Temple	Kumbakonam Temple Cluster	road_transport	6	70	[1]
Gangaikondacholapuram Temple	Mayuranathaswami Temple, Mayiladuthurai	road_transport	5	60	[5]
Nidur Archaeological Remains	Sri Rajagopalaswamy Temple	road_transport	4	8	[2] [5]
Papanasam Shiva Temple	Kumbakonam Temple Cluster	road_transport	5	25	[5]
Papanasam Shiva Temple	Brihadeeswarar Temple, Thanjavur	road_transport	5	45	[5]
Cauvery Delta Wetlands	Point Calimere Wildlife Sanctuary	ecological	7	50	[4] [5]
Swaminatha Temple, Swamimalai	Sikkal Singaravelan Temple	thematic_pilgrimage	6	60	[1]
Vedaranyeswarar Temple, Vedaranyam	Thiruthuraipoondi Market Town	road_transport	5	30	[5]
Karaikal French Heritage Area	Nagapattinam Heritage Area	road_transport	6	15	[5]

6.3 Edge Reason Annotations

The following table provides the full documented reason for each edge connection:

From	To	Documented Reason
Sri Rajagopalaswamy Temple	Sakkara Kulam (Temple Tank)	Temple tank physically adjacent; integral to rituals
Sri Rajagopalaswamy Temple	Mannargudi Old Town & Market	Old market agraharam streets surround the temple complex
Mannargudi Old Town & Market	Sakkara Kulam (Temple Tank)	Market streets border the tank; festivals draw pilgrims to both
Sri Rajagopalaswamy Temple	Thyagarajaswamy Temple, Thiruvarur	Regular bus on SH66; major Thiruvarur district pilgrimage centres
Thyagarajaswamy Temple, Thiruvarur	Thiruvarur Temple Chariot Museum	Chariot museum immediately adjacent to the Thyagaraja complex
Thyagarajaswamy Temple, Thiruvarur	Thirukkaravasal Shiva Temple	Both Paadal Petra Sthalams on Thiruvarur Shaiva circuit
Thirukkaravasal Shiva Temple	Sri Rajagopalaswamy Temple	~5 km from Mannargudi; pilgrims combine visits in single day
Thyagarajaswamy Temple, Thiruvarur	Agniswarar Temple, Nannilam	Both Paadal Petra Sthalams in Thiruvarur district pilgrimage circuit
Agniswarar Temple, Nannilam	Sikkal Singaravelan Temple	~12 km via local roads; combined Shaiva-Murugan pilgrimage route
Sri Rajagopalaswamy Temple	Cauvery Delta Wetlands	Mannargudi lies within the Cauvery delta; wetlands surround the town
Cauvery Delta Wetlands	Thiruthuraipoondi Market Town	Thiruthuraipoondi is gateway town to coastal delta; bus services run
Thiruthuraipoondi Market Town	Point Calimere Wildlife Sanctuary	Point Calimere accessed via Thiruthuraipoondi / Vedaranyam road
Point Calimere Wildlife Sanctuary	Vedaranyam Salt Marshes	Adjacent coastal zones; same day eco-tourism visit is common
Vedaranyam Salt Marshes	Vedaranyeswarar Temple, Vedaranyam	Vedaranyeswarar temple is in same town as the salt marshes

Point Calimere Wildlife Sanctuary	Kodikkarai Lighthouse	Kodikkarai lighthouse is at Point Calimere itself
Sri Rajagopalaswamy Temple	Darbaranyeswarar Temple, Thirunallar	~60 km via Nagapattinam; organised pilgrim buses run this route
Darbaranyeswarar Temple, Thirunallar	Karaikal French Heritage Area	Thirunallar is ~5 km from Karaikal; most visitors combine both
Nagapattinam Heritage Area	Sikkal Singaravelan Temple	Sikkal is ~20 km from Nagapattinam; standard day trip for visitors
Nagapattinam Heritage Area	Darbaranyeswarar Temple, Thirunallar	~15 km apart; Karaikal-Nagapattinam pilgrim corridor
Sri Rajagopalaswamy Temple	Nagapattinam Heritage Area	~45 km via SH66; regular bus connectivity
Sri Rajagopalaswamy Temple	Kumbakonam Temple Cluster	~55 km via NH83; Kumbakonam major hub with frequent buses
Kumbakonam Temple Cluster	Airavatesvara Temple, Darasuram	Darasuram 4 km from Kumbakonam; UNESCO site visited as day extension
Kumbakonam Temple Cluster	Swaminatha Temple, Swamimalai	Swamimalai is 5 km from Kumbakonam; part of standard Kumbakonam tour
Kumbakonam Temple Cluster	Brihadeeswarar Temple, Thanjavur	~38 km via NH83; Thanjavur-Kumbakonam most visited Chola heritage corridor
Sri Rajagopalaswamy Temple	Brihadeeswarar Temple, Thanjavur	~65 km via NH83; standard day trip for tourists based in Mannargudi
Brihadeeswarar Temple, Thanjavur	Airavatesvara Temple, Darasuram	Both UNESCO Great Living Chola Temples; marketed together by TN Tourism
Mayuranathaswami Temple, Mayiladuthurai	Thyagarajaswamy Temple, Thiruvarur	~35 km apart; both major Shaiva temples on Kaveri-delta pilgrimage route
Mayuranathaswami Temple, Mayiladuthurai	Kumbakonam Temple Cluster	~30 km; both on NH83 / Kaveri temple circuit
Mayuranathaswami Temple, Mayiladuthurai	Nagapattinam Heritage Area	~25 km; both on the Cauvery coast pilgrimage circuit

Gangaikondacholapuram Temple	Kumbakonam Temple Cluster	~70 km; both Chola heritage sites; Chola circuit by TN Tourism
Gangaikondacholapuram Temple	Mayuranathaswami Temple, Mayiladuthurai	~60 km; pilgrimage route north Kaveri delta to Mayiladuthurai
Nidur Archaeological Remains	Sri Rajagopalaswamy Temple	~8 km from Mannargudi; accessible by local auto-rickshaw
Papanasam Shiva Temple	Kumbakonam Temple Cluster	~25 km from Kumbakonam via NH83; part of Cauvery Shaiva circuit
Papanasam Shiva Temple	Brihadeeswarar Temple, Thanjavur	~45 km; day trip possible from Thanjavur
Cauvery Delta Wetlands	Point Calimere Wildlife Sanctuary	Both Ramsar wetland sites in Cauvery delta ecological zone
Swaminatha Temple, Swamimalai	Sikkal Singaravelan Temple	Both Arupadaiveedu Murugan shrines; pilgrims often combine visits
Vedaranyeswarar Temple, Vedaranyam	Thiruthuraipoondi Market Town	~30 km; Thiruthuraipoondi transit hub for Vedaranyam-bound pilgrims
Karaikal French Heritage Area	Nagapattinam Heritage Area	~15 km; Karaikal and Nagapattinam coastal heritage cluster

7. SNA Metrics & Key Findings

7.1 Graph-Level Statistics

Metric	Value
Total Nodes (|V|)	25
Total Edges (|E|)	38
Graph Density	0.1267
Is Connected	Yes (single component)
Average Degree	3.04
Average Clustering Coefficient	0.268
Number of Communities	4
Modularity (Q)	0.528

7.2 Top-10 Nodes by Betweenness Centrality

Betweenness centrality identifies sites that act as indispensable gateways in tourist flow routing. A site with high betweenness controls the most shortest paths between all pairs of sites — its removal would most severely disrupt connectivity.

Rk	Site	Degree	Betw.	Close.	Eig.	PageR.	Comm.
1	**Sri Rajagopalaswamy Temple**	0.3333	**0.6697**	0.5625	1.0000	0.0897	0
2	**Cauvery Delta Wetlands**	0.1250	0.3442	0.3966	0.1841	0.0310	0
3	**Nagapattinam Heritage Area**	0.1667	0.1812	0.3714	0.2793	0.0384	1
4	**Cauvery Delta → Point Calimere**	0.1250	0.1755	0.3633	0.1601	0.0264	0
5	**Kumbakonam Temple Cluster**	0.2083	0.1630	0.4222	0.5913	0.0632	2
6	**Mayuranathaswami Temple**	0.1250	0.1289	0.3807	0.3629	0.0406	2
7	**Thyagarajaswamy Temple, Thiruvarur**	0.1667	0.1202	0.3617	0.3265	0.0435	0
8	**Brihadeeswarar Temple, Thanjavur**	0.1250	0.0721	0.3541	0.4048	0.0468	2
9	**Thiruthuraipoondi Market Town**	0.0833	0.0651	0.3137	0.0868	0.0197	0

10	**Thirunallar Temple**	0.1250	0.0495	0.3315	0.1895	0.0279	1

7.3 Community Structure

Four communities were detected with a modularity of Q = 0.528, indicating strongly separated sub-circuits:

ID	Circuit Name	Key Sites & Thematic Description
0	**Core Mannargudi Circuit**	Sri Rajagopalaswamy Temple, Sakkara Kulam, Mannargudi Market, Thirukkaravasal Temple, Thyagarajaswamy Temple (Thiruvarur), Nannilam Temple, Cauvery Delta Wetlands, Thiruthuraipoondi — the primary religious and cultural hub of the Thiruvarur District.
1	**Southern Coastal Arc**	Nagapattinam Heritage Area, Thirunallar Darbaranyeswarar Temple, Karaikal French Heritage Area, Sikkal Singaravelan Temple — coastal pilgrimage and heritage sites south of Mannargudi along SH66.
2	**Chola Heritage Belt**	Kumbakonam Temple Cluster, Swamimalai Temple, Darasuram Airavatesvara Temple (UNESCO), Brihadeeswarar Temple Thanjavur (UNESCO), Papanasam Temple, Gangaikondacholapuram (UNESCO), Mayuranathaswami Temple Mayiladuthurai — the premier Chola-period architectural corridor.
3	**Eco & Coastal South**	Point Calimere Wildlife Sanctuary (Ramsar), Vedaranyam Salt Marshes, Vedaranyam Temple, Kodikkarai Lighthouse, Nidur Archaeological Site — eco-tourism and coastal heritage at the southern tip of the Thiruvarur district.

7.4 Key Analytical Findings

9. Sri Rajagopalaswamy Temple is the dominant gateway: betweenness centrality 0.670 (2× the next highest node). Any effective tourism dispersion strategy must route tourists through or from this site.
10. The Cauvery Delta Wetlands (betweenness 0.344) function as a structural bridge between the religious pilgrimage community and the eco-tourism community, making it the single most important node for inter-circuit visitor flow.
11. Three UNESCO-listed Chola temples (Brihadeeswarar, Darasuram, Gangaikondacholapuram) cluster together in Community 2 with high eigenvector centrality (0.40–0.59), indicating mutual reinforcement — visitors to one are highly likely to visit the others.
12. Nidur Archaeological Site and Kodikkarai Lighthouse are structural periphery nodes (betweenness < 0.01) representing under-visited heritage opportunities, accessible within 8 km of major hubs.
13. The Kumbakonam cluster shows the highest strength score (42), reflecting dense, high-weight connections — it is the most intensely connected sub-cluster in the network.

8. Interest Tag Taxonomy

The itinerary generation system maps user interest categories to node-level tags. Each node carries a list of fine-grained tags; user interest categories map to tag sets as follows:

User Interest Category	Mapped Tags (node attribute match)
Spiritual / Pilgrimage	vishnu, shiva, murugan, vaishnava, shaiva, paadal_petra, arupadaiveedu, pilgrimage, chola, ritual, theertham
Chola Heritage & Architecture	chola, unesco, world_heritage, sculpture, architecture, nayak
Eco-Tourism & Wildlife	ramsar, wetland, birdwatching, wildlife, flamingo, blackbuck, delta, coastal
History & Archaeology	colonial, archaeology, jain, medieval, french, dutch, port, lighthouse, satyagraha
Culture & Craft	market, silk, craft, bronze, food, museum, chariot, culture

9. Application Features

9.1 Six Interactive Tabs

Tab	Description
Network Map	Spring-layout graph coloured by node type. Node size proportional to betweenness centrality. Edge width proportional to connection weight. Hover tooltips show full site metadata.
Geographic View	Plotly Scattergeo map using actual GPS coordinates. Shortest-path finder: user selects source and destination, path is highlighted in red, hop count and estimated km displayed. Node size proportional to PageRank.
Metrics Dashboard	Switchable centrality bar charts (5 measures). Scatter plot: betweenness vs. annual visitors. Full sortable centrality table with conditional colour formatting. Metric definitions panel.
Community Detection	Spring-layout graph coloured by Louvain/greedy community assignment. Modularity score displayed. Per-community breakdown tables showing all member sites and their tags.
Itinerary Builder	Sidebar inputs (interests, days, start node) → network-ranked personalised itinerary. Day-by-day card view with site descriptions and gateway scores. Interactive flow diagram showing the itinerary path overlaid on the full network. Estimated travel distance table.
Site Explorer	Select any of 25 sites. Radar chart of 5 centrality metrics. Full metadata display. Connected sites table with edge details. Complete shortest path table to all reachable nodes.

10. Data Sources & References

Ref	Source	Used For
[1]	**Tamil Nadu Tourism Dept. — tamilnadutourism.tn.gov.in**	Site classification, visitor estimates, circuit connections, pilgrimage routes
[2]	**Archaeological Survey of India — Thiruvarur District Survey Reports**	Node historical periods, Nidur archaeological site description
[3]	**HR&CE Dept. Tamil Nadu — tnhrce.gov.in**	Paadal Petra Sthalam classification (Shaiva sacred sites list)
[4]	**Wikipedia — Mannargudi, Rajagopalaswamy Temple, Thiruvarur, Point Calimere etc.**	GPS coordinates cross-check, descriptions, historical context
[5]	**Google Maps / OpenStreetMap (osm.org)**	GPS coordinates, road distances, route verification
[6]	**Lonely Planet India — Tamil Nadu Chapter**	Tourism circuit connections, site significance ratings
[7]	**Temples of Tamil Nadu — S. R. Balasubramanian (published documentation)**	Temple historical periods, Chola-era construction details
[8]	**Thiruvarur District Collectorate Tourism Records**	Local market sites, cultural attractions, visitor flow estimates

11. Limitations

14. Visitor estimates are approximate figures from district tourism reports and news articles, not official census-level data. Actual footfall may vary seasonally by ±40%.
15. Edge weights are researcher-assigned based on qualitative factors. Measured tourist flow data (e.g., transport ticketing records) would produce a more accurate weighted graph.
16. The graph is static and does not model seasonal variation: monsoon road closures, festival-period surges (Panguni Uthiram at Mannargudi draws 500,000+ visitors in a single week), or infrastructure changes.
17. The itinerary algorithm is heuristic and does not account for opening hours, entry fees, physical accessibility constraints, or travel time within a day.
18. The network covers sites within approximately 90 km of Mannargudi. Important regional sites beyond this radius (Chidambaram, Rameswaram, Thiruvannamalai) are excluded.
19. Community detection uses a non-deterministic greedy algorithm; results may vary with different random seeds, though the fixed seed (42) ensures reproducibility.
20. No user tracking or survey data was collected. The visitor_annual attribute is a supply-side estimate (reported capacity/registration), not demand-side measurement.

12. Ethics Statement

12.1 Data Ethics

- No personal data of any kind is collected, stored, or processed. The application runs entirely locally without network calls.
- All data is drawn from publicly available, non-confidential sources. No private records, internal temple documents, or restricted ASI surveys were used.
- Visitor estimates are aggregate figures from public tourism reports and do not identify individuals.

12.2 Cultural Sensitivity

- Temple sites are represented through publicly available historical and geographical information only. No internal religious texts, ritual protocols, or private ceremonial information is described or reproduced.
- All temples are described in terms of their architectural heritage, historical period, and geographical connectivity — not in terms of religious doctrine or belief systems.
- The Vedaranyam Salt Satyagraha (1930) historical connection is documented factually as part of the site's public heritage, consistent with standard Indian history curriculum.

12.3 Research Integrity

- Every node, edge, and data attribute is cited to its source. Users can independently verify any claim using the reference list in Section 10.
- All limitations (Section 11) are disclosed transparently in both this report and the application footer.
- Random seeds are fixed (42) to ensure full reproducibility of community detection results.

- No data was invented, extrapolated beyond its documented basis, or used to make claims beyond what the network structure can support.

12.4 Community Benefit

- The primary motivation for this research is equitable tourism dispersion — identifying under-visited but culturally significant sites as candidates for targeted promotion, potentially reducing over-crowding at major sites like the Rajagopalaswamy Temple.
- The application is freely deployable and the codebase is open for audit and extension by Tamil Nadu tourism planners, local government, and researchers.

13. Deployment Instructions

13.1 Prerequisites

- Python 3.9 or higher
- pip (Python package manager)
- 4 GB RAM minimum; 8 GB recommended
- Internet connection for initial package installation only

13.2 Installation Steps

```
# Step 1: Unzip the project
unzip mannargudi_sna.zip && cd mannargudi_sna

# Step 2: Create and activate a virtual environment (recommended)
python -m venv venv
source venv/bin/activate        # macOS / Linux
venv\Scripts\activate.bat       # Windows

# Step 3: Install dependencies
pip install -r requirements.txt

# Note: python-louvain is optional — the app falls back to
# NetworkX's built-in greedy modularity if unavailable.

# Step 4: Launch the Streamlit app
streamlit run app.py

# App opens at http://localhost:8501
```

13.3 Dependency Resolution Note

The python-louvain package (community detection library) may not be available on all pip mirrors. The codebase includes an automatic fallback to NetworkX's built-in greedy modularity communities algorithm, which produces equivalent results without any additional installation. No action is required from the user — the fallback activates silently if import fails.

14. Future Work

21. Incorporate real tourist flow data from transport ticketing (TNSTC bus ridership) to replace researcher-assigned edge weights with empirically measured values.
22. Add temporal modelling: seasonal edge weight variation based on festival calendars and monsoon road condition data.
23. Extend the network beyond 90 km to include Chidambaram, Madurai, and Rameswaram, enabling cross-district SNA at the state level.
24. Deploy as a public web application (Streamlit Cloud or Hugging Face Spaces) with an updated dataset API feed from Tamil Nadu Tourism Department.
25. Conduct surveys of actual tourists at the Rajagopalaswamy Temple to validate network centrality predictions against real visitation patterns.
26. Apply link prediction algorithms (Jaccard coefficient, Adamic-Adar index) to identify potential new tourism connections worth developing (e.g., new bus routes or guided tour circuits).

27. Integrate with Google Maps API to provide real-time travel duration and public transport schedule data within the itinerary generator.

End of Report

Mannargudi Tourism SNA · Thiruvarur District, Tamil Nadu, India · All data publicly verifiable